

Harry Nguyen

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EDUCATION

Georgia State University

Expected May 2027

Bachelor of Science in Computer Science • **GPA: 3.75/4.0**

- Coursework: Distributed Systems, Operating Systems, Database Systems, Computer Networks, Data Structures & Algorithms
- Honors: Presidential Scholarship, Anita Andrew Fellowship, Roxie Alexander Memorial Award

WORK EXPERIENCE

Google | Software Engineer Intern — Chrome Browser

May 2025 – Aug 2025

- Designed **C++ IPC transport layer** with Protocol Buffers serving **3B+ active users** at **sub-50ms p99, 10K+ req/sec** — selected Protobuf over custom serialization for schema evolution and cross-team compatibility at global scale
- Profiled settings navigation as p99 bottleneck at 1,200ms; implemented **lock-free concurrent trie search** eliminating mutex contention — **96% latency reduction**, zero production regressions in Chrome stable channel
- Architected event-driven TypeScript/React system with observer pattern across **25K+ lines** of Chromium at **95% test coverage** — design documents adopted into production branch by senior Chrome engineers

CoderPush | Software Engineer Intern

Sep 2023 – Dec 2023

- Diagnosed DynamoDB hot-partition failure at **9,000+ req/sec**; redesigned partition key strategy — **30% read throughput improvement** under sustained high-concurrency traffic
- Designed idempotent payment APIs with Redis distributed caching (**85% cache hit rate**) and exponential backoff — enforced exactly-once semantics preventing duplicate transactions under network partitions

Develop for Good | Software Engineer Intern

May 2024 – Aug 2024

- Designed stateless BaaS on AWS — selected JWT over session auth for horizontal scalability; auto-scaling supporting **500+ concurrent users**, **90% deployment time reduction** via CI/CD (GitHub Actions)
- Diagnosed N+1 query bottleneck degrading response to 3s+ on large datasets; redesigned PostgreSQL indexing achieving **sub-100ms** for 10,000+ records

PERSONAL PROJECTS

TiMoto AI — ML Serving Platform | *vLLM, gRPC, Django, PostgreSQL, AWS, Terraform* timoto.ai

- **Sole engineer** — designed, built, and operate end-to-end platform: vLLM inference engine, gRPC inter-service layer, Django REST backend, PostgreSQL, and multi-AZ AWS cloud infrastructure
- Resolved **production deadlock under concurrent gRPC calls** — traced root cause to circular resource acquisition; redesigned call sequencing — **100% success rate** at **sub-50ms p99**
- Architected multi-AZ ECS Fargate with Terraform IaC and **circuit breaker pattern** — **99.9% uptime, 44% cost reduction** to \$40–60/month; auto-rollback on health check failure
- Selected **vLLM with PagedAttention** over naive inference to eliminate KV cache memory fragmentation under concurrent load — zero OOM failures at production traffic

Pulumi — Open Source Contributor | *Go, TypeScript, IaC*

24.4K★ github.com/pulumi/pulumi

- Submitted Go CLI features and bug fixes enabling multi-cloud (AWS/Azure/GCP) provisioning — under active review by core maintainers
- Analyzed **Raft/Paxos consensus** in distributed state synchronization layer — studied linearizability guarantees and correctness under concurrent operations and partial failures

TECHNICAL SKILLS

Distributed Systems: gRPC, Protocol Buffers, circuit breakers, exactly-once semantics, linearizability, Raft/Paxos, lock-free concurrency, event-driven architecture, fault tolerance

Performance Engineering: p99 latency profiling, lock-free data structures, mutex elimination, hot-partition diagnosis, sub-50ms SLOs, throughput optimization

Languages: C++, Python, Go, TypeScript, Java, JavaScript, SQL, Bash, Rust

Cloud & Infrastructure: AWS (ECS Fargate, EC2, S3, RDS, DynamoDB), Docker, Kubernetes, Terraform, CI/CD, GitHub Actions

Frameworks & Databases: Django, FastAPI, Node.js, React, REST APIs, PostgreSQL, Redis, MongoDB